ProtCOVER: Family-Aware Conformal Prediction for Variant Effect Predictors on Unseen Protein Families

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Abstract. Protein language models and related variant effect predictors (VEPs) are increasingly used to classify missense variants, yet their reliability is typically summarized by benchmark-wide averages. In practice, predictors are applied to proteins that are evolutionarily distant from those used for calibration, which constitutes a family-level distribution shift. Standard split conformal prediction assumes exchangeable variants and offers no guarantee when entire protein families are unseen, while existing VEP calibration schemes condition on residue, gene or domain attributes rather than on homology support. We present ProtCOVER, a conformal procedure that treats homology families, rather than individual variants, as the exchangeable unit. ProtCOVER (i) constructs label-free homology clusters and calibrates at the family level, yielding marginal coverage for a new family under family exchangeability; (ii) reweights calibration families using estimated family-level density ratios from label-free descriptors (sequence similarity, alignment depth and embedding distance) to adapt to descriptor shift; and (iii) computes the effective calibration support for each query protein and returns a conservative fallback set, or abstains, when support is insufficient. We state the assumptions under which each guarantee holds and separately measure empirical robustness where they do not. We evaluate ProtCOVER on the ProteinGym clinical substitution benchmark using precomputed zero-shot scores from five predictors under leave-family-out splits, against split, Mondrian, group-weighted and hierarchical conformal baselines. Class-conditional split conformal, calibrated over variants, covers only 0.749–0.790 of pathogenic variants per family for four of the five predictors at a 0.90 target, even without shift. Under descriptor shift, family-level calibration alone falls to 0.798, while ProtCOVER keeps family-weighted pathogenic and benign coverage at or above 0.892 for every predictor and setting. The support gate preserves these guarantees, but on this benchmark support-based abstention did not reduce the error of retained predictions relative to random abstention. Code is publicly available.

Keywords: conformal prediction · variant effect prediction · protein language models · distribution shift · selective prediction · protein families

1 Introduction

Zero-shot variant effect predictors (VEPs), from protein language models such as ESM-1b [28,5] to alignment-based models such as EVE and GEMME [10,14] and hybrids such as TranceptEVE and PoET [23,32], now rank missense variants with high accuracy on clinical benchmarks [22]. Clinical use, however, needs more than a ranking: a variant interpretation framework such as ACMG/AMP [27] needs evidence whose error rate is known, and the ClinGen calibration of computational evidence [25] has shown that a single genome-wide calibration can hide large differences between genes [30].

Conformal prediction [34,1] offers distribution-free error control: a prediction set over $\{\text{benign}, \text{pathogenic}\}$ that contains the true label with probability at least $1 - \alpha$. Its guarantee rests on exchangeability between calibration and test data. For VEPs this assumption fails in two ways. First, variants are grouped into proteins and homologous protein families: a predictor is accurate or inaccurate for a whole family at once, and families contribute very different numbers of labeled variants (from 2 to 741 per protein in our data). Pooling variants lets the largest families set the error rate, so the "90%" is a statement about variants from well-studied genes, not about a new protein. Second, the proteins a predictor is deployed on are often evolutionarily distant from, or have shallower alignments than, the proteins in the calibration set: a family-level distribution shift that is visible without labels.

We make three contributions.

- We formalize *family-weighted* coverage, the probability that a random variant of a new protein family is covered, and its class-conditional version for pathogenic variants. On ProteinGym clinical substitutions, class-conditional (Mondrian) split conformal, the standard remedy for class imbalance, meets its own variant-level target yet covers only 0.749–0.790 of pathogenic variants per family for four of five predictors at a 0.90 target, with no distribution shift at all.
- We present ProtCOVER, which (i) calibrates at the family level, (ii) reweights calibration families by a cross-fitted density ratio of label-free family descriptors, and (iii) gates queries with insufficient calibration support. We prove finite-sample family-weighted coverage under family exchangeability (Proposition 1, a restatement of hierarchical conformal prediction [16]) and under family-level covariate shift (Theorem 1), bound the loss from estimated weights (Corollary 1), and show that the gate preserves all of these (Lemma 1).
- We evaluate five predictors under exchangeable, synthetic-shift, and orphan-family splits with family-level bootstrap intervals. Under shift, ProtCOVER keeps both class-conditional coverages at or above 0.892 for every predictor and setting, against 0.798 for family-level calibration alone and 0.873 for group-weighted conformal prediction. We also report a negative result: abstaining on poorly supported families did not reduce the error of the remaining predictions relative to random abstention.

2 Related Work

Conformal prediction. Split conformal prediction [34,17] gives marginal coverage under exchangeability; Mondrian or class-conditional versions calibrate separately within categories [33,8]. Weighted conformal prediction handles covariate shift with a known or estimated likelihood ratio [31], with coverage loss bounded by the ratio's estimation error [18]; related reweightings address label shift [26] and general departures from exchangeability [2]. Group-weighted conformal prediction [3] sharpens these bounds when the shift acts through a finite set of known groups; we use it with descriptor bins as a baseline. Localized conformal prediction [13] reweights calibration points by proximity to the query, and robust and multi-source methods protect against families of target distributions [6,20]. For grouped data, Dunn et al. [9] proposed pooling and subsampling methods for a new group, and Lee et al. [16] proved finite-sample guarantees for the pooled construction (HCP); later work uses a few observations from the test group [21]. Macro-coverage [4] averages coverage over classes rather than over samples, in the same spirit as our family weighting. ProtCOVER combines family-level pooling with family-level density-ratio weights, which none of these works does, and applies it to protein families defined by homology.

Calibration of variant effect predictors. ClinGen calibrated computational scores to evidence strengths for PP3/BP4 [25]; subsequent work found that genome-wide calibration masks heterogeneous performance across genes [30] and proposed gene- and domain-aware calibration [7], and RaCoon calibrates ESM-1b at the residue level with conditional score distributions [24]. These methods improve calibration of point probabilities within subgroups seen in training; they do not give distribution-free guarantees, and they condition on gene, domain or residue attributes rather than on homology support relative to the calibration data. Kermut [12] provides calibrated uncertainty for supervised variant effect regression on deep mutational scans. ProteinGym [22] supplies the benchmark and precomputed scores we use, including those of PLM-based and alignment-based predictors [28,10,14,23,32] and of supervised clinical predictors, which we exclude because they are trained on ClinVar labels [15]. Selective classification [11] motivates our risk-coverage evaluation of abstention.

3 Problem Setup

Families and scores. A protein *family* is a cluster of homologous wild-type sequences obtained without labels (Section 6.1). Family k is the random object $G_k = (D_k, N_k, (X_{k,i}, Y_{k,i})_{i=1}^{N_k})$, where $X_{k,i}$ describes a missense variant, $Y_{k,i} \in \{0, 1\}$ is its label (1 = pathogenic), and $D_k \in \mathcal{D}$ is a *label-free descriptor* of the family: any function of its sequences, alignments, predictor scores, and of the fixed set of training families, but not of labels. A predictor score is turned into $\hat{p}(y \mid x)$ by a base model fitted on separate training families, which we treat as fixed. The nonconformity score is $s(x, y) = 1 - \hat{p}(y \mid x)$ and $S_{k,i} = s(X_{k,i}, Y_{k,i})$. We observe calibration families $G_1, \ldots, G_K$ with labels and a test family G_{K+1} whose descriptor D_{K+1} and variants $X_{K+1,i}$ are observed but whose labels are not.

Coverage targets. For a set-valued predictor $C : \mathcal{X} \to 2^{\{0,1\}}$, the per-family coverage of the test family is $\gamma_{K+1}(C) = N_{K+1}^{-1} \sum_i \mathbf{1}\{Y_{K+1,i} \in C(X_{K+1,i})\}$. Our primary target is *family-weighted coverage*

$$\mathrm{Cov}(C) = \mathbb{E}\big[\gamma_{K+1}(C)\big], \tag{1}$$

the probability that a uniformly random variant of a new family is covered; every family counts once, whatever its number of variants. Its class-conditional version for label y is $\mathrm{Cov}_y(C) = \mathbb{E}\big[N_{K+1,y}^{-1} \sum_{i:Y_{K+1,i}=y} \mathbf{1}\{y \in C(X_{K+1,i})\} \,\big|\, N_{K+1,y} > 0\big]$, where $N_{k,y}$ counts label-y variants of family k. For clinical use, Cov_1 (pathogenic coverage) is the quantity of interest: $1 - \mathrm{Cov}_1$ is the family-weighted rate at which a pathogenic variant is confidently called benign. Variant-weighted coverage, which pools all test variants, lets families with many variants dominate; we report it but do not target it.

Assumptions. We use two alternative assumptions; each result states which one it needs.

A1 (*Family exchangeability.*) $(G_1, \ldots, G_{K+1})$ is exchangeable.

A2 (*Family-level covariate shift.*) $G_1, \ldots, G_K \overset{\text{iid}}{\sim} P$ and, independently, $G_{K+1} \sim Q$, where $Q \ll P$ and $\frac{dQ}{dP}(G) = w(D)$ for a measurable $w : \mathcal{D} \to [0, \infty)$. Equivalently, the conditional law of a family's variants and labels given its descriptor is the same under P and Q; only the distribution of descriptors shifts.

A1 holds when calibration and test families are drawn by a random split of a pool of families, regardless of dependence between families. It does not require exchangeability of variants within a family, because Cov averages over all variants of the test family (this is weaker than the hierarchical exchangeability of [16], which targets one variant of the new group). A2 formalizes deployment on families that differ from the calibration families in label-free ways, such as being evolutionarily distant or having shallow alignments.

4 ProtCOVER

(i) Family-level calibration. Let $\hat{F}_k = N_k^{-1} \sum_i \delta_{S_{k,i}}$ be the empirical distribution of calibration scores of family k. Given nonnegative family weights $\omega_1, \ldots, \omega_{K+1}$, set $p_k = \omega_k / \sum_{j=1}^{K+1} \omega_j$ and

$$\hat{\mu} = \sum_{k=1}^{K} p_k \hat{F}_k + p_{K+1}\, \delta_{+\infty}, \qquad \hat{q} = \mathrm{Q}_{1-\alpha}(\hat{\mu}), \qquad C(x) = \{y \in \{0,1\} : s(x,y) \le \hat{q}\}, \tag{2}$$

where $\mathrm{Q}_\tau(\mu) = \inf\{t : \mu((-\infty, t]) \ge \tau\}$. With $\omega \equiv 1$ each calibration family has total weight $1/(K+1)$ spread evenly over its variants, and the test family's mass sits at $+\infty$; this is the hierarchical conformal prediction (HCP) threshold of [16], which equals the CDF-pooling method of [9] up to the level adjustment. Variant-level split conformal instead gives every calibration variant weight 1, so families with hundreds of variants dominate the quantile.

(ii) Family-level weighting. To adapt to descriptor shift, ProtCOVER sets $\omega_k = \hat{w}(D_k)$ for an estimate $\hat{w}$ of the density ratio $w = dQ/dP$ on descriptors, and $\omega_{K+1} = \hat{w}(D_{K+1})$; the test family's mass is computable because D_{K+1} is label-free. The estimate comes from a logistic classifier separating calibration families (class 0) from target families (class 1) on standardized descriptors: $\hat{w}(d) = \frac{n_0}{n_1} \exp\{\hat{f}(d)\}$ with $\hat{f}$ the fitted log-odds, L_2 regularization chosen by five-fold cross-validated log loss on this membership task (no variant labels), weights rescaled to mean one over the calibration families used for the quantile, and clipped to $[1/c, c]$ ($c = 20$). We use descriptors for sequence similarity (maximum identity of the family to any training protein, over significant MMseqs2 hits), alignment depth ($\log N_{\text{eff}}$ and N_{eff} per position), and embedding distance (cosine distance of mean-pooled ESM2 embeddings to the nearest training protein), together with log length and log number of scored variants.

Cross-fitting. The guarantee for estimated weights (Corollary 1) needs $\hat{w}$ to be independent of the families it weights. We therefore split the calibration families once, at random, into a third used only to fit $\hat{w}$ and two thirds used in (2), and partition the target families into five folds; the ratio applied to fold j (and to the calibration families, for the queries of fold j) is fitted against the target families outside fold j.

(iii) Support gate. For a query family t with standardized descriptor x_t, the localized calibration weights $\kappa_{tk} = \exp(-\|x_k - x_t\|^2/2h^2)$ give the effective calibration support

$$\mathrm{ESS}_t = \Big(\sum_k \kappa_{tk}\Big)^2 \Big/ \sum_k \kappa_{tk}^2, \tag{3}$$

the effective number of calibration families that resemble the query; the test-point mass $\pi_t = p_{K+1}$ measures how atypical the query is under the weights of (ii). The gate returns the full set $\{0, 1\}$ (abstains) when $\mathrm{ESS}_t < m$ or $\pi_t > \tau$. The bandwidth h (a fraction of the median pairwise descriptor distance), m, and τ are tuned only on an inner split of the training families (Section 6.2).

(iv) Class-conditional versions. Each component has a Mondrian version: for each candidate label y, the threshold $\hat{q}_y$ is computed from (2) using only label-y calibration variants ($\hat{F}_k^y$ over the $N_{k,y}$ label-y variants, families with $N_{k,y} = 0$ dropped), and $y \in C(x)$ iff $s(x, y) \le \hat{q}_y$. These target Cov_0 and Cov_1 separately.

5 Guarantees

All results are finite-sample and distribution-free given the stated assumption; proofs are in Appendix A. They hold for any fixed score function, so they do not depend on the quality of the base model or of the predictor.

Proposition 1 (Family-level calibration). *Under A1, the set* (2) *with* $\omega \equiv 1$ *satisfies* $\mathrm{Cov}(C) \ge 1 - \alpha$.

Proposition 1 is Theorem 1 of [16] with the test point replaced by a uniformly random variant of the test family; we include the short proof because it removes the need for within-family exchangeability. If scores are almost surely distinct, [16] also give the upper bound $1 - \alpha + 2/(K + 1)$ for their target.

Theorem 1 (Family-level weighted calibration). *Under A2, the set* (2) *with* $\omega_k = w(D_k)$, $k = 1, \ldots, K+1$, *satisfies* $\mathrm{Cov}(C) \ge 1 - \alpha$, *where the expectation is over* $G_1, \ldots, G_K \sim P$ *and* $G_{K+1} \sim Q$.

Corollary 1 (Estimated weights). *Under A2, let* $\hat{w}$ *be independent of* $(G_1, \ldots, G_{K+1})$ *with* $0 < \mathbb{E}_P \hat{w}(D) < \infty$, *and let* $\hat{Q}$ *be defined by* $d\hat{Q}/dP = \hat{w}(D)/\mathbb{E}_P \hat{w}(D)$. *With* $\omega_k = \hat{w}(D_k)$,

$$\mathrm{Cov}(C) \;\ge\; 1 - \alpha - d_{\mathrm{TV}}(Q, \hat{Q}) \;=\; 1 - \alpha - \tfrac{1}{2}\,\mathbb{E}_P\Big|w(D) - \frac{\hat{w}(D)}{\mathbb{E}_P \hat{w}(D)}\Big|.$$

The argument is that of [18] (their Theorem 3) lifted to families. Under cross-fitting, for queries in fold j the fitted ratio is independent of the calibration families used in (2) and of the query family, so the corollary applies fold by fold. Clipping and rescaling produce another fixed function of D, to which the corollary applies directly.

Proposition 2 (Class-conditional versions). *The Mondrian versions of Proposition 1, Theorem 1, and Corollary 1 hold with* Cov *replaced by* Cov_y, *for each* $y \in \{0, 1\}$, *provided* $\mathbb{P}(N_{K+1,y} > 0) > 0$.

Lemma 1 (Enlargement). *If* $C'(x) \supseteq C(x)$ *for all* x, *almost surely, where* C' *may depend on all observed data, then* $\mathrm{Cov}(C') \ge \mathrm{Cov}(C)$ *and* $\mathrm{Cov}_y(C') \ge \mathrm{Cov}_y(C)$.

Because the gate (iii) only replaces a set by $\{0, 1\}$, Lemma 1 shows that it preserves every guarantee above, whatever its parameters.

What is and is not guaranteed. The guarantees are marginal over families. They do not imply coverage for a particular family, nor coverage on the queries the gate retains (a selected subset). Corollary 1 bounds the loss from estimated weights by a quantity that is not observable. Our S2 splits sample families without replacement from a finite pool, and S3 has no explicit shift model, so A2 holds only approximately there. Everything we report about per-family dispersion, retained queries, S2, and S3 is therefore empirical; Section 6.4 checks the guarantees themselves on synthetic data where A1 or A2 hold exactly.

6 Experiments

6.1 Data, predictors, and families

We use the ProteinGym v1.3 clinical substitution benchmark [22]: 62,727 ClinVar missense variants in 2,525 human proteins, 51.0% pathogenic, with a median of 12 labeled variants per protein (maximum 741). We take ProteinGym's precomputed zero-shot scores for five unsupervised predictors: ESM-1b (protein language model), EVE and GEMME (alignment-based), TranceptEVE (hybrid), and PoET (retrieval-augmented language model). We orient every score so that higher means more pathogenic, using ProteinGym's directionality settings; our per-protein mean AUCs reproduce ProteinGym's published values (0.892–0.920) with a maximum absolute difference of 0.000. Supervised clinical predictors in the same files are excluded because they are trained on ClinVar.

Families are MMseqs2 [29] clusters of wild-type sequences at 30% identity and 80% bidirectional coverage: 2,081 families, of which 1,805 are single proteins (largest: 10 proteins) and 1,309 have at least ten variants. As sensitivity checks we use 50% identity (2,275 families) and single-linkage components of significant homology ($E \leq 10^{-3}$, identity $\geq 30\%$, both sequences covered $\geq 50\%$; 1,939 families), which merges homologs that share a domain. Family descriptors are computed without labels: maximum identity to any training protein over significant MMseqs2 hits, $\log N_{\text{eff}}$ and N_{eff} per position from ProteinGym's alignment weights, cosine distance of mean-pooled ESM2-35M [19] embeddings to the nearest training protein, log length, log number of scored variants, and, for one ablation, summaries of the predictor's scores. A family is an *orphan* if no member has a significant hit at $\geq 30\%$ identity outside the family (1,276 families).

6.2 Protocol

Each of ten seeds splits families into 40% training, 30% calibration and 30% test (832, 624 and 625 families). **S1** draws all three uniformly at random. **S2** keeps the S1 training families and samples test families without replacement with probability proportional to $\exp(-\beta z)$, where z is the standardized mean of maximum identity to training and $\log N_{\text{eff}}$, so that test families are poorly supported and calibration families well supported; $\beta \in \{0.5, 1, 2\}$ ($\beta = 0$ is S1). At $\beta = 1$, mean maximum identity to training is 0.105 for test and 0.221 for calibration families, and mean $\log N_{\text{eff}}$ is 5.88 against 6.93. **S3** draws test families only from orphan families, a natural shift with no explicit model (mean maximum identity 0.048 against 0.236).

The base model, fitted on training families only, is a logistic regression on the raw score (clipped to the training 0.1% and 99.9% quantiles) and the score standardized within its protein; its test AUC is 0.873–0.929 across predictors. Baselines are variant-level split conformal (B1) and its class-conditional version (B2); family-level HCP pooling and Dunn et al.'s single and repeated subsampling (B3) [16,9]; group-weighted conformal (B4) [3] with nine bins (maximum identity zero, low or high, crossed with $\log N_{\text{eff}}$ terciles) and target proportions from the test families; and support-only and random abstention at ProtCOVER's abstention rate (B5). Every method has marginal and class-conditional versions. ProtCOVER uses 416 of the calibration families in the quantile and the rest to fit $\hat{w}$. The gate's kernel scale is chosen by inner risk-coverage AUC and (m, τ) as the least-abstaining setting within 0.005 of the best retained coverage under a pre-declared 10% abstention budget, all on a shifted ($\beta = 1$) inner split of the training families. Test families are used only for final evaluation, and only their label-free descriptors enter $\hat{w}$.

We report family-weighted coverage Cov, class-conditional Cov_1 (pathogenic) and Cov_0 (benign), variant-weighted coverage, mean set size, singleton rate, the fraction of families with at least ten variants whose coverage is below $1 - \alpha - 0.05$ (with its expectation under pure binomial noise), abstention rates, and risk-coverage curves (error of retained singleton sets against the retained fraction, abstaining whole families) with their area (AURC). Intervals are 95% bootstrap intervals that resample test families within each seed and average over seeds; differences between methods use the same resamples (paired). Unless stated otherwise $\alpha = 0.1$; results at $\alpha \in \{0.05, 0.2\}$ are in Appendix B.

6.3 Results

Variant-level calibration misses pathogenic variants per family. In S1 (Fig. 1), variant-level split conformal has variant-weighted coverage 0.894–0.901, close to its target, but family-weighted coverage 0.880–0.885;

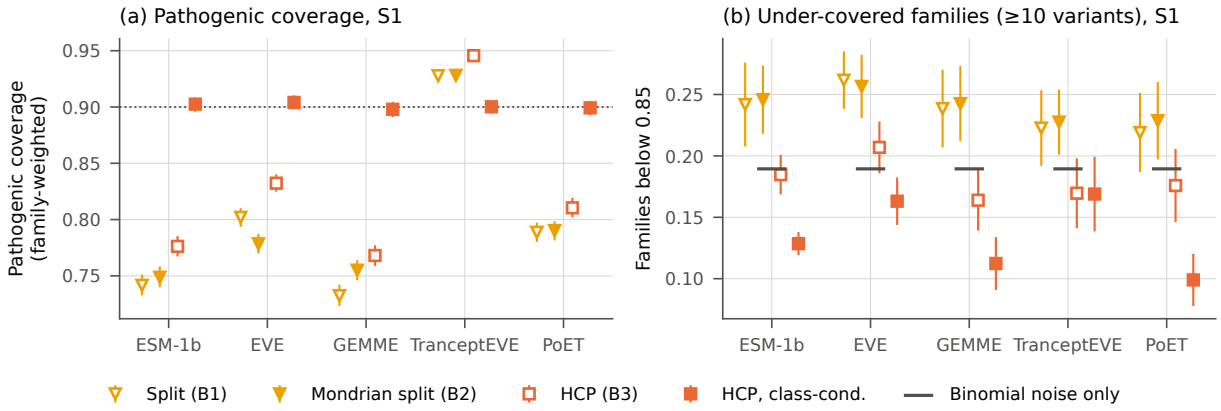


Fig. 1. Exchangeable families (S1, $\alpha = 0.1$). (a) Family-weighted pathogenic coverage Cov_1 with 95% family-bootstrap intervals; the dotted line is the target. (b) Fraction of families with at least ten variants whose coverage is below 0.85 (mean and s.d. over seeds); the gray dash is the expected fraction if every family had coverage exactly 0.90.

Table 1. Family-weighted pathogenic / benign coverage (Cov_1 / Cov_0) of class-conditional methods at $\alpha = 0.1$ (target 0.90), mean over ten seeds. Paired differences with intervals are in Table 3.

Setting	Method	ESM-1b	EVE	GEMME	TranceptEVE	PoET
S1 (exchangeable)	Mondrian split (B2)	0.749 / 0.899	0.778 / 0.898	0.755 / 0.895	0.928 / 0.837	0.790 / 0.877
	HCP (B3)	0.903 / 0.903	0.904 / 0.901	0.898 / 0.898	0.900 / 0.901	0.899 / 0.903
	GWCP (B4)	0.902 / 0.901	0.902 / 0.899	0.897 / 0.897	0.900 / 0.900	0.899 / 0.901
	ProtCOVER	**0.897 / 0.904**	**0.904 / 0.902**	**0.895 / 0.899**	**0.902 / 0.902**	**0.897 / 0.906**
S2, $\beta = 1$	Mondrian split (B2)	0.655 / 0.929	0.709 / 0.915	0.712 / 0.906	0.920 / 0.839	0.795 / 0.856
	HCP (B3)	0.838 / 0.927	0.867 / 0.916	0.879 / 0.909	0.884 / 0.894	0.908 / 0.875
	GWCP (B4)	0.899 / 0.904	0.888 / 0.909	0.891 / 0.904	0.895 / 0.894	0.891 / 0.897
	ProtCOVER	**0.910 / 0.905**	**0.895 / 0.909**	**0.902 / 0.908**	**0.897 / 0.898**	**0.897 / 0.905**
S2, $\beta = 2$	Mondrian split (B2)	0.619 / 0.941	0.686 / 0.917	0.700 / 0.910	0.921 / 0.839	0.806 / 0.840
	HCP (B3)	0.798 / 0.939	0.855 / 0.917	0.869 / 0.914	0.880 / 0.891	0.920 / 0.853
	GWCP (B4)	0.888 / 0.905	0.877 / 0.906	0.887 / 0.897	0.888 / 0.889	0.895 / 0.873
	ProtCOVER	**0.902 / 0.915**	**0.896 / 0.916**	**0.898 / 0.915**	**0.908 / 0.900**	**0.910 / 0.892**
S3 (orphan families)	Mondrian split (B2)	0.678 / 0.915	0.772 / 0.895	0.724 / 0.903	0.943 / 0.823	0.817 / 0.854
	HCP (B3)	0.876 / 0.908	0.909 / 0.899	0.890 / 0.905	0.914 / 0.892	0.931 / 0.879
	GWCP (B4)	0.889 / 0.905	0.903 / 0.898	0.895 / 0.900	0.903 / 0.913	0.901 / 0.900
	ProtCOVER	**0.894 / 0.908**	**0.907 / 0.901**	**0.900 / 0.908**	**0.900 / 0.921**	**0.901 / 0.910**

family-level HCP restores 0.900–0.901. The gap is concentrated in the pathogenic class: B2, calibrated class by class over variants, attains variant-weighted pathogenic coverage 0.893–0.905 for four predictors yet covers only 0.749–0.790 of pathogenic variants per family. The 5% of test families with the most pathogenic variants hold 50% of them, and their pathogenic variants are easier: under B2, families with fewer than five pathogenic variants have pathogenic coverage 0.660–0.727 against 0.899–0.921 for families with at least twenty. TranceptEVE is the exception (0.928), with the error moved to the benign class. Family-level class-conditional calibration attains 0.898–0.904 (pathogenic) and 0.898–0.903 (benign), as Propositions 1 and 2 guarantee, and lowers the fraction of under-covered families below the binomial-noise level (Fig. 1b), at the cost of larger sets.

Weighting restores coverage under family-level shift. Under S2 (Fig. 2, Table 1), family-level calibration alone degrades as β grows: at $\beta = 1$ its pathogenic coverage spans 0.838–0.908 across predictors, and for ESM-1b it falls from 0.838 at $\beta = 1$ to 0.798 at $\beta = 2$. ProtCOVER's weighting keeps pathogenic coverage at 0.895–0.910 ($\beta = 1$) and 0.896–0.910 ($\beta = 2$), and benign coverage at 0.898–0.909 and 0.892–0.916; for ESM-1b the paired gain over HCP is +0.072 [+0.067, +0.078] at $\beta = 1$ and +0.104 [+0.098, +0.111] at $\beta = 2$. Group-weighted conformal, whose bins use the same two descriptors that define the S2 shift, is the strongest baseline, yet ProtCOVER's pathogenic coverage exceeds it by +0.002 to +0.012 at $\beta = 1$ (paired interval above zero for 5 of five predictors) and by +0.011 to +0.020 at $\beta = 2$ (5 of five). On orphan families (S3) the shift is

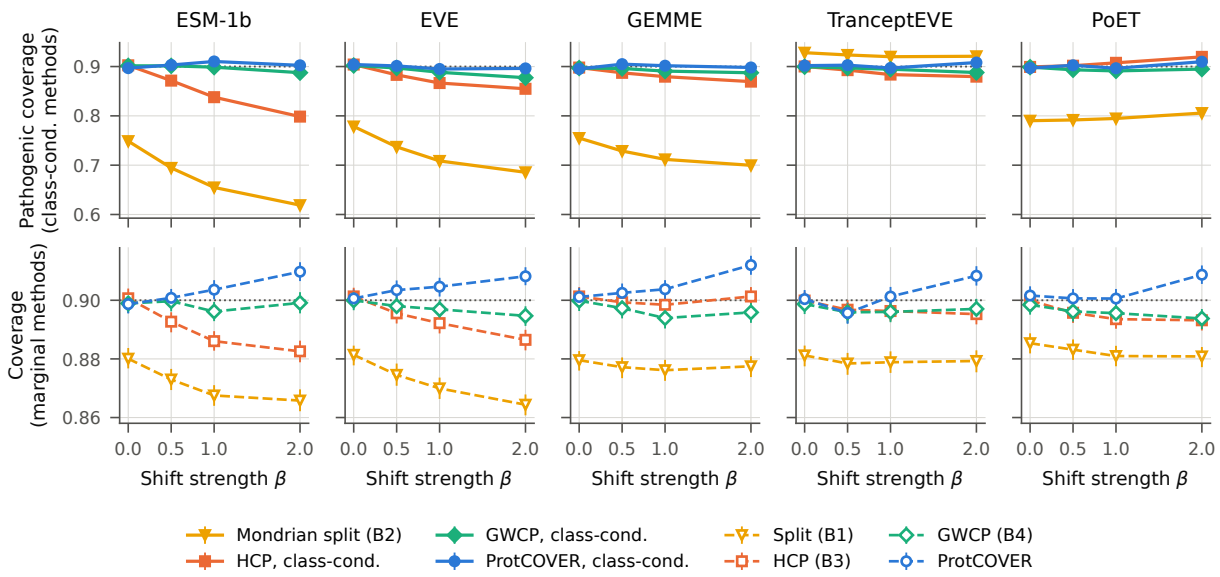


Fig. 2. Coverage under increasing descriptor shift (S2; $\beta = 0$ is S1), $\alpha = 0.1$, with 95% family-bootstrap intervals. Top: pathogenic coverage Cov_1 of class-conditional methods. Bottom: family-weighted coverage Cov of marginal methods.

milder: ProtCOVER's pathogenic coverage spans 0.894–0.907 against 0.876–0.931 for HCP, whose error is not one-sided (it over-covers pathogenic variants for PoET and TranceptEVE while under-covering their benign variants; Table 1). Across all shifted settings, predictors and both classes, the worst coverage is 0.892 for ProtCOVER, 0.873 for GWCP, and 0.798 for HCP. The price is larger sets (Fig. 3): at $\beta = 1$, mean set size is 1.290–1.428 for ProtCOVER against 1.260–1.329 for HCP and 1.269–1.397 for GWCP. Along the α curves, ProtCOVER's curve lies above HCP's for ESM-1b and EVE under S2; elsewhere the family-level curves nearly coincide, so the extra size mostly moves along a shared coverage-size trade-off toward the target rather than improving it.

The support gate preserves coverage but does not improve selective risk. With (m, τ) and the kernel scale tuned on training families under a 10% budget, the gate abstained on at most 1.0% of test families except at $\beta = 2$ (0.0%–5.5%), so the full method matches (i)+(ii) (Table 2). Fig. 4 evaluates the ranking itself at every abstention level. Abstaining on families with the lowest ESS gave a lower AURC than random abstention in the majority of seeds in only 9 of 25 predictor-setting cells (higher in 14); ranking by maximum identity alone did no better (7 better, 13 worse), nor did the test-point mass (13 better, 11 worse). On this benchmark, how well a family is supported by calibration data does not predict how often the predictor errs on it once coverage is controlled; the gate's value is as a guard against weight degeneracy, which Lemma 1 makes free in terms of coverage.

Ablations. Table 2 shows that sequence similarity or embedding distance alone does not correct the S2 shift, alignment depth alone helps on average but not for every predictor, and the support descriptors together match the full set; adding the predictor's score summaries does not help. Embedding distance contributes little here because it barely differs between calibration and test families. Tight clipping ($c = 5$) under-corrects at $\beta = 2$. Under the alternative family definitions, ProtCOVER's mean pathogenic coverage over predictors stays at 0.901–0.906 across the shifted settings; its worst predictor-setting cell is 0.873 with 50% clustering and 0.893 with homology components, against 0.788 and 0.804 for HCP.

6.4 Synthetic validation

We simulate families with a descriptor $d \sim N(0, 1)$, random offsets, and slopes on the score whose mean depends on d, with a fixed base model, so that the truth is known (300 replicates, Monte Carlo standard error

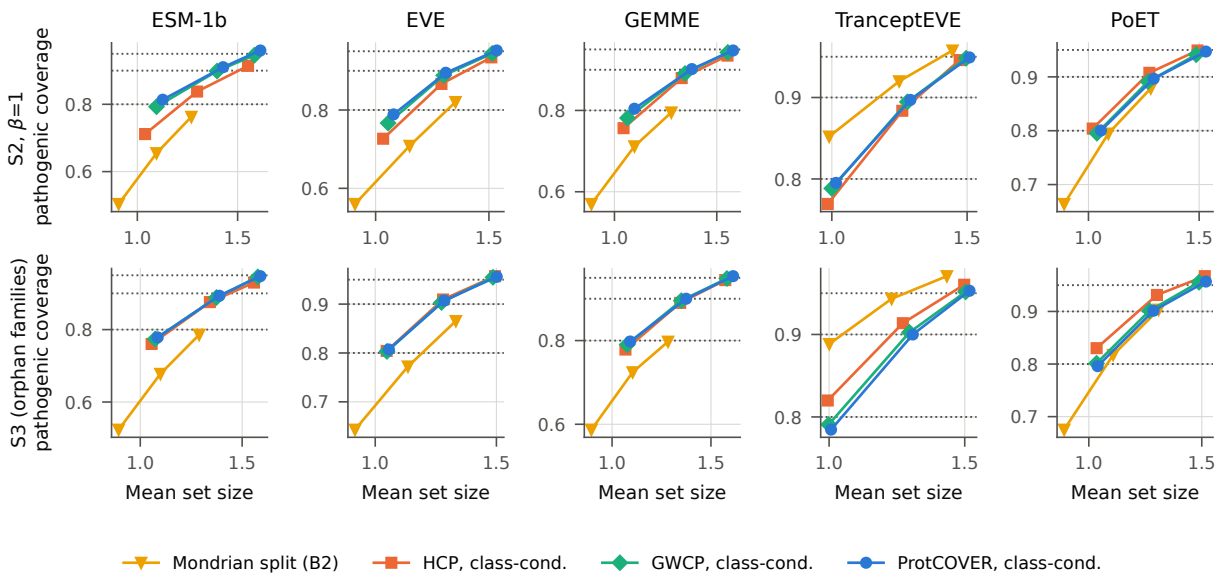


Fig. 3. Pathogenic coverage against mean set size for $\alpha \in \{0.2, 0.1, 0.05\}$ (left to right along each curve), class-conditional methods. Dotted lines mark the three targets.

at most 0.001). Under A1 with larger families being easier, variant-level split conformal has variant-weighted coverage 0.900 but family-weighted coverage 0.831, while HCP attains 0.906 and single subsampling 0.902. Under exact family-level covariate shift (A2, test descriptors $N(-\beta, 1)$), HCP drops to 0.826 at $\beta = 1$ and 0.734 at $\beta = 2$; weighting with the true ratio attains 0.911 and 0.950, and with cross-fitted estimated ratios 0.917 and 0.956 (class-conditional pathogenic: 0.916 and 0.953). These checks are also unit tests in the released code.

7 Limitations

The guarantees are marginal over families: ProtCOVER does not certify coverage for any particular protein, and the per-family coverage of small families is dominated by binomial noise. The weighted guarantee assumes that, given its descriptors, a family's labels and scores follow the same law in calibration and deployment; our S2 splits satisfy this only approximately (finite-pool sampling), S3 not by construction, and Corollary 1 bounds the effect of estimated weights by an unobservable distance. The 30%/80% clustering leaves most families as single proteins, and partial-coverage homologs can fall in different families; the stricter homology components did not change our conclusions, but deeper remote homology could still link training and test families. ClinVar labels are noisy and enriched in well-studied genes, and every protein in the benchmark has both pathogenic and benign variants, which is not true of a typical query protein. The base model is deliberately simple, and set sizes would shrink with a stronger one. Finally, the support gate did not improve selective risk here; whether support predicts error for other predictors, variant classes, or benchmarks remains open.

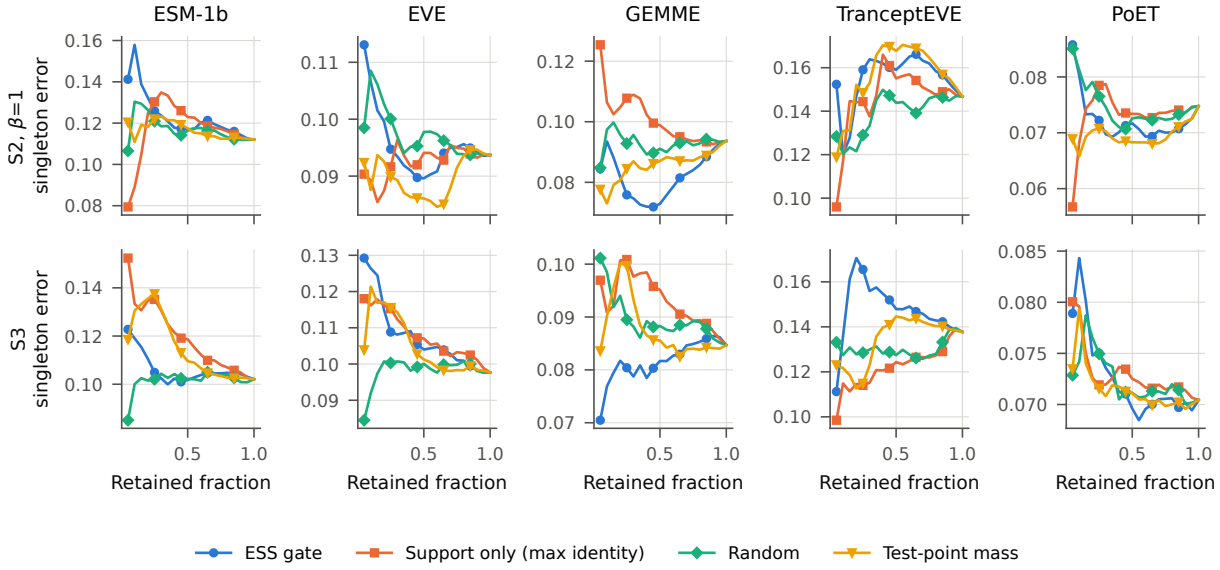


Fig. 4. Risk-coverage of abstention rules applied to ProtCOVER's class-conditional sets ($\alpha = 0.1$, mean over seeds): error of retained singleton sets as whole families are abstained in order of the score.

Table 2. Ablations: pathogenic coverage Cov_1 at $\alpha = 0.1$, mean (minimum) over the five predictors. For family definitions, HCP / ProtCOVER.

Variant	S2, $\beta = 1$	S2, $\beta = 2$	S3
Components			
(i) family-level calibration (HCP)	0.875 (0.838)	0.864 (0.798)	0.904 (0.876)
(i)+(ii) weighting	0.900 (0.895)	0.903 (0.896)	0.900 (0.894)
(i)+(iii) gate	0.875 (0.838)	0.864 (0.798)	0.904 (0.876)
(i)+(ii)+(iii) full	0.900 (0.895)	0.905 (0.898)	0.901 (0.894)
Descriptors for $\hat{w}$			
sequence similarity only	0.871 (0.850)	0.853 (0.799)	0.899 (0.892)
alignment depth only	0.899 (0.892)	0.901 (0.872)	0.899 (0.878)
embedding distance only	0.879 (0.845)	0.866 (0.806)	0.904 (0.872)
the three support groups	0.900 (0.894)	0.904 (0.898)	0.900 (0.893)
main set + predictor score summaries	0.900 (0.885)	0.891 (0.869)	0.901 (0.895)
Weight clipping c (default 20)			
$c = 5$	0.899 (0.894)	0.890 (0.874)	0.901 (0.892)
$c = 100$	0.900 (0.895)	0.908 (0.904)	0.901 (0.894)
no clipping	0.900 (0.895)	0.908 (0.904)	0.901 (0.894)
Family definition (HCP / ProtCOVER)			
MMseqs2 30% id., 80% cov. (main)	0.875 (0.838) / 0.900 (0.895)	0.864 (0.798) / 0.903 (0.896)	0.904 (0.876) / 0.900 (0.894)
MMseqs2 50% id., 80% cov.	0.882 (0.835) / 0.903 (0.897)	0.869 (0.788) / 0.903 (0.873)	0.903 (0.872) / 0.906 (0.897)
homology components	0.881 (0.839) / 0.901 (0.896)	0.871 (0.804) / 0.902 (0.893)	0.910 (0.894) / 0.905 (0.901)

Code Availability

Code, configuration, and scripts that regenerate every number, table, and figure in this paper from the public ProteinGym release are available at `https://github.com/wael-fahmy/protcover` under the MIT license.

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A Proofs

Throughout, $F_k = N_k^{-1} \sum_{i=1}^{N_k} \delta_{S_{k,i}}$ is the score distribution of family k (for the test family it involves unobserved labels and is used only in the analysis), and $\gamma_{K+1} = F_{K+1}((-\infty, \hat{q}])$ because a test variant is covered iff its score is at most $\hat{q}$.

A comparison step used in every proof. Fix weights $p_1, \dots, p_{K+1}$ and define the oracle measure $\mu = \sum_{k=1}^{K+1} p_k F_k$ and $q = \mathrm{Q}_{1-\alpha}(\mu)$. The computable $\hat{\mu}$ of (2) is obtained from μ by moving the mass p_{K+1} from F_{K+1} to $+\infty$, so $\hat{\mu}((-\infty, t]) \le \mu((-\infty, t])$ for every t and hence $\hat{q} \ge q$. Consequently $\gamma_{K+1} = F_{K+1}((-\infty, \hat{q}]) \ge F_{K+1}((-\infty, q])$. Also $\mu((-\infty, q]) \ge 1 - \alpha$ by right-continuity of distribution functions.

Proof (Proposition 1). With $\omega \equiv 1$, $p_k = 1/(K+1)$ and q is a symmetric function of $(G_1, \dots, G_{K+1})$. By exchangeability, $\mathbb{E}[F_{K+1}((-\infty, q])] = \mathbb{E}[F_k((-\infty, q])]$ for every k, so

$$\mathbb{E}\big[F_{K+1}((-\infty, q])\big] = \mathbb{E}\Big[\frac{1}{K+1} \sum_{k=1}^{K+1} F_k((-\infty, q])\Big] = \mathbb{E}\big[\mu((-\infty, q])\big] \ge 1 - \alpha.$$

The comparison step gives $\mathrm{Cov}(C) = \mathbb{E}\gamma_{K+1} \ge 1 - \alpha$.

Proof (Theorem 1). Under A2 the joint law of $(G_1, \dots, G_{K+1})$ has density $\prod_{k \le K} f(g_k) \cdot w(d_{K+1}) f(g_{K+1})$ with respect to a product base measure, where f is a density of P; the sequence is therefore weighted exchangeable with weight functions $1, \dots, 1, w$ in the sense of [31, Definition 1 and Lemma 2]. Let E be the event that the unordered collection $\{G_1, \dots, G_{K+1}\}$ equals $\{g_1, \dots, g_{K+1}\}$. Exactly as in the proof of Lemma 3 of [31], $\mathbb{P}(G_{K+1} = g_k \mid E) = w(d_k)/\sum_j w(d_j) = p_k$ (summing p_k over coincident values, if any). The weights p_k and hence μ and q are functions of E alone, so

$$\mathbb{E}\big[F_{K+1}((-\infty, q]) \,\big|\, E\big] = \sum_{k=1}^{K+1} p_k \, F_k((-\infty, q]) = \mu((-\infty, q]) \ge 1 - \alpha.$$

Taking expectations and applying the comparison step yields $\mathrm{Cov}(C) \geq 1-\alpha$. The weights in (2) are computable because they depend on the test family only through the label-free D_{K+1}.

Proof (Corollary 1). Condition on $\hat{w}$, which is independent of the families. The procedure is invariant to rescaling the weights, so it coincides with the procedure of Theorem 1 for the shift $\hat{Q}$, whose density ratio with respect to P is $\hat{w}/\mathbb{E}_P\hat{w}$. Let $h(g) \in [0,1]$ be the per-family coverage of a test family g, given $\hat{w}$ and $G_1, \ldots, G_K$ (the threshold depends on g only through its descriptor). Theorem 1 gives $\mathbb{E}[\mathbb{E}_{\hat{Q}}h] \geq 1-\alpha$, and $|\mathbb{E}_Q h - \mathbb{E}_{\hat{Q}}h| \leq d_{\mathrm{TV}}(Q,\hat{Q})$ for any $[0,1]$-valued h. Hence $\mathrm{Cov}(C) = \mathbb{E}[\mathbb{E}_Q h] \geq 1-\alpha-d_{\mathrm{TV}}(Q,\hat{Q})$, and $d_{\mathrm{TV}}(Q,\hat{Q}) = \frac{1}{2}\mathbb{E}_P|w(D) - \hat{w}(D)/\mathbb{E}_P\hat{w}(D)|$.

Proof (Proposition 2). Fix y and let $\mathcal{K}_y = \{k \leq K+1 : N_{k,y} > 0\}$. Under A1, conditionally on $\mathcal{K}_y$ and on $K+1 \in \mathcal{K}_y$, the families indexed by $\mathcal{K}_y$ are exchangeable, because the event $\{\mathcal{K}_y = \kappa\}$ is invariant under permutations of κ. Applying the proof of Proposition 1 to these families, with F_k replaced by the label-y score distribution F_k^y, gives $\mathrm{Cov}_y(C) \geq 1-\alpha$. Under A2, conditioning each family on $N_{\cdot,y} > 0$ keeps the families independent, with laws $P_y = P(\cdot \mid N_y > 0)$ and $Q_y = Q(\cdot \mid N_y > 0)$ satisfying $dQ_y/dP_y \propto w(D)$ because $\mathbf{1}\{N_y > 0\}$ is a function of G. The proof of Theorem 1 then applies verbatim, and the normalizing constant cancels in p_k. The corollary follows in the same way.

Proof (Lemma 1). $\mathbf{1}\{Y \in C'(X)\} \geq \mathbf{1}\{Y \in C(X)\}$ pointwise, so every average and expectation of the indicators is at least as large for C'.

B Additional Results

Table 3. Paired differences in family-weighted pathogenic coverage Cov_1 between class-conditional methods at $\alpha = 0.1$, with 95% paired family-bootstrap intervals.

Setting	Comparison	ESM-1b	EVE	GEMME	TranceptEVE	PoET
S1 (exchangeable)	ProtCOVER − HCP	-0.006 [-0.008, -0.004]	+0.000 [-0.001, +0.001]	-0.003 [-0.004, -0.001]	+0.002 [+0.001, +0.003]	-0.002 [-0.004, -0.001]
	ProtCOVER − GWCP	-0.005 [-0.006, -0.003]	+0.002 [+0.001, +0.003]	-0.002 [-0.003, +0.000]	+0.002 [+0.001, +0.003]	-0.002 [-0.003, -0.000]
	HCP − Mondrian split	+0.154 [+0.147, +0.161]	+0.126 [+0.119, +0.132]	+0.143 [+0.136, +0.150]	-0.028 [-0.031, -0.025]	+0.109 [+0.103, +0.115]
S2, $\beta = 1$	ProtCOVER − HCP	+0.072 [+0.067, +0.078]	+0.028 [+0.025, +0.032]	+0.022 [+0.019, +0.025]	+0.013 [+0.011, +0.016]	-0.011 [-0.013, -0.009]
	ProtCOVER − GWCP	+0.012 [+0.009, +0.014]	+0.007 [+0.005, +0.009]	+0.011 [+0.009, +0.013]	+0.002 [+0.001, +0.004]	+0.005 [+0.003, +0.008]
	HCP − Mondrian split	+0.183 [+0.176, +0.190]	+0.158 [+0.150, +0.166]	+0.168 [+0.161, +0.175]	-0.036 [-0.040, -0.033]	+0.113 [+0.107, +0.119]
S2, $\beta = 2$	ProtCOVER − HCP	+0.104 [+0.098, +0.111]	+0.041 [+0.037, +0.045]	+0.029 [+0.025, +0.033]	+0.029 [+0.026, +0.031]	-0.010 [-0.012, -0.007]
	ProtCOVER − GWCP	+0.015 [+0.011, +0.019]	+0.019 [+0.015, +0.022]	+0.011 [+0.008, +0.014]	+0.020 [+0.017, +0.023]	+0.015 [+0.012, +0.018]
	HCP − Mondrian split	+0.180 [+0.172, +0.187]	+0.169 [+0.161, +0.177]	+0.170 [+0.162, +0.178]	-0.041 [-0.045, -0.038]	+0.114 [+0.107, +0.120]
S3 (orphan families)	ProtCOVER − HCP	+0.018 [+0.015, +0.020]	-0.002 [-0.004, -0.000]	+0.010 [+0.008, +0.012]	-0.014 [-0.016, -0.012]	-0.030 [-0.033, -0.027]
	ProtCOVER − GWCP	+0.005 [+0.003, +0.007]	+0.004 [+0.002, +0.006]	+0.005 [+0.003, +0.006]	-0.003 [-0.005, -0.001]	-0.000 [-0.002, +0.002]
	HCP − Mondrian split	+0.198 [+0.190, +0.207]	+0.137 [+0.130, +0.144]	+0.166 [+0.158, +0.173]	-0.030 [-0.032, -0.027]	+0.114 [+0.108, +0.120]

Tables 4–6 report every method, marginal (cc = no) and class-conditional (cc = yes), averaged over the five predictors: family-weighted coverage Cov, variant-weighted coverage, Cov_1, Cov_0, mean set size, singleton rate, the fraction of families with at least ten variants whose coverage is below $1-\alpha-0.05$, and the fraction of test families on which the method abstained. Per-predictor values with bootstrap intervals are in the files `results/main/summary.csv` and `results/main/seed_summary.csv` of the code repository.

Table 4. All methods at $\alpha = 0.1$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.882	0.899	0.798	0.884	1.143	0.857	0.237	0.000
	Split (B1/B2)	yes	0.881	0.898	0.800	0.881	1.142	0.858	0.240	0.000
	HCP (B3)	no	0.901	0.918	0.827	0.903	1.204	0.796	0.180	0.000
	HCP (B3)	yes	0.922	0.929	0.901	0.901	1.304	0.696	0.134	0.000
	Subsample once (B3)	no	0.900	0.918	0.828	0.902	1.204	0.796	0.180	0.000
	Subsample once (B3)	yes	0.922	0.929	0.901	0.900	1.305	0.695	0.136	0.000
	Repeated subsample (B3)	no	0.901	0.918	0.826	0.903	1.205	0.795	0.180	0.000
	Repeated subsample (B3)	yes	0.923	0.929	0.901	0.901	1.305	0.695	0.134	0.000
	GWCP (B4)	no	0.899	0.917	0.824	0.901	1.199	0.801	0.185	0.000
	GWCP (B4)	yes	0.921	0.928	0.900	0.899	1.300	0.700	0.136	0.000
	ProtCOVER (i)+(ii)	no	0.900	0.918	0.826	0.903	1.204	0.796	0.181	0.000
	ProtCOVER (i)+(ii)	yes	0.923	0.930	0.899	0.902	1.303	0.697	0.133	0.000
	ProtCOVER (i)+(iii)	no	0.901	0.918	0.827	0.903	1.204	0.796	0.180	0.000
	ProtCOVER (i)+(iii)	yes	0.923	0.929	0.901	0.901	1.304	0.696	0.134	0.000
	ProtCOVER full	no	0.901	0.918	0.826	0.903	1.204	0.796	0.180	0.000
	ProtCOVER full	yes	0.923	0.930	0.899	0.903	1.304	0.696	0.133	0.000
	Support-only abst. (B5)	no	0.901	0.918	0.826	0.903	1.204	0.796	0.181	0.000
	Support-only abst. (B5)	yes	0.923	0.930	0.899	0.902	1.304	0.696	0.133	0.000
	Random abst. (B5)	no	0.901	0.918	0.826	0.903	1.204	0.796	0.180	0.000
	Random abst. (B5)	yes	0.923	0.930	0.899	0.902	1.304	0.696	0.133	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.875	0.893	0.772	0.883	1.137	0.863	0.252	0.000
	Split (B1/B2)	yes	0.873	0.891	0.758	0.889	1.137	0.863	0.255	0.000
	HCP (B3)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	HCP (B3)	yes	0.917	0.925	0.875	0.904	1.291	0.709	0.147	0.000
	Subsample once (B3)	no	0.893	0.912	0.800	0.902	1.195	0.805	0.200	0.000
	Subsample once (B3)	yes	0.918	0.926	0.877	0.904	1.294	0.706	0.146	0.000
	Repeated subsample (B3)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	Repeated subsample (B3)	yes	0.917	0.925	0.875	0.904	1.290	0.710	0.147	0.000
	GWCP (B4)	no	0.896	0.914	0.805	0.903	1.202	0.798	0.195	0.000
	GWCP (B4)	yes	0.921	0.928	0.893	0.902	1.318	0.682	0.140	0.000
	ProtCOVER (i)+(ii)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	ProtCOVER (i)+(ii)	yes	0.925	0.931	0.900	0.905	1.340	0.660	0.130	0.000
	ProtCOVER (i)+(iii)	no	0.893	0.912	0.800	0.901	1.195	0.805	0.200	0.000
	ProtCOVER (i)+(iii)	yes	0.917	0.925	0.875	0.904	1.291	0.709	0.147	0.000
	ProtCOVER full	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	ProtCOVER full	yes	0.925	0.931	0.900	0.905	1.340	0.660	0.129	0.000
	Support-only abst. (B5)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	Support-only abst. (B5)	yes	0.925	0.931	0.900	0.905	1.341	0.659	0.130	0.000
	Random abst. (B5)	no	0.903	0.921	0.818	0.909	1.227	0.773	0.176	0.000
	Random abst. (B5)	yes	0.925	0.931	0.900	0.905	1.341	0.659	0.130	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.874	0.892	0.764	0.883	1.136	0.864	0.255	0.000
	Split (B1/B2)	yes	0.871	0.890	0.746	0.889	1.136	0.864	0.259	0.000
	HCP (B3)	no	0.892	0.911	0.793	0.900	1.192	0.808	0.204	0.000
	HCP (B3)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	Subsample once (B3)	no	0.889	0.909	0.792	0.896	1.187	0.813	0.211	0.000
	Subsample once (B3)	yes	0.912	0.922	0.863	0.901	1.279	0.721	0.159	0.000
	Repeated subsample (B3)	no	0.891	0.911	0.792	0.900	1.192	0.808	0.204	0.000
	Repeated subsample (B3)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	GWCP (B4)	no	0.896	0.915	0.801	0.904	1.212	0.788	0.196	0.000
	GWCP (B4)	yes	0.915	0.923	0.887	0.894	1.315	0.685	0.159	0.000
	ProtCOVER (i)+(ii)	no	0.909	0.928	0.827	0.915	1.264	0.736	0.159	0.000
	ProtCOVER (i)+(ii)	yes	0.927	0.935	0.903	0.908	1.365	0.635	0.126	0.000
	ProtCOVER (i)+(iii)	no	0.892	0.911	0.793	0.900	1.192	0.808	0.204	0.000
	ProtCOVER (i)+(iii)	yes	0.914	0.923	0.864	0.903	1.283	0.717	0.156	0.000
	ProtCOVER full	no	0.909	0.928	0.827	0.915	1.264	0.736	0.159	0.000
	ProtCOVER full	yes	0.928	0.936	0.905	0.908	1.374	0.626	0.125	0.022
	Support-only abst. (B5)	no	0.909	0.928	0.827	0.915	1.264	0.736	0.159	0.000
	Support-only abst. (B5)	yes	0.928	0.936	0.905	0.909	1.376	0.624	0.125	0.022
	Random abst. (B5)	no	0.909	0.928	0.827	0.915	1.264	0.736	0.159	0.000
	Random abst. (B5)	yes	0.928	0.936	0.905	0.909	1.377	0.623	0.124	0.022
S3 (orphan families)	Split (B1/B2)	no	0.875	0.891	0.789	0.878	1.137	0.863	0.254	0.000
	Split (B1/B2)	yes	0.874	0.890	0.787	0.878	1.135	0.865	0.251	0.000
	HCP (B3)	no	0.897	0.913	0.821	0.899	1.204	0.796	0.193	0.000
	HCP (B3)	yes	0.922	0.931	0.904	0.896	1.308	0.692	0.129	0.000
	Subsample once (B3)	no	0.896	0.912	0.820	0.898	1.204	0.796	0.195	0.000
	Subsample once (B3)	yes	0.922	0.931	0.904	0.896	1.309	0.691	0.128	0.000
	Repeated subsample (B3)	no	0.896	0.912	0.821	0.899	1.203	0.797	0.194	0.000
	Repeated subsample (B3)	yes	0.922	0.931	0.904	0.896	1.308	0.692	0.129	0.000
	GWCP (B4)	no	0.901	0.917	0.827	0.903	1.220	0.780	0.182	0.000
	GWCP (B4)	yes	0.924	0.931	0.898	0.903	1.312	0.688	0.128	0.000
	ProtCOVER (i)+(ii)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	ProtCOVER (i)+(ii)	yes	0.928	0.934	0.900	0.910	1.329	0.671	0.120	0.000
	ProtCOVER (i)+(iii)	no	0.897	0.913	0.821	0.899	1.204	0.796	0.193	0.000
	ProtCOVER (i)+(iii)	yes	0.922	0.931	0.904	0.897	1.309	0.691	0.129	0.003
	ProtCOVER full	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	ProtCOVER full	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004
	Support-only abst. (B5)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	Support-only abst. (B5)	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004
	Random abst. (B5)	no	0.905	0.922	0.834	0.908	1.238	0.762	0.170	0.000
	Random abst. (B5)	yes	0.928	0.934	0.901	0.910	1.331	0.669	0.120	0.004

Table 5. All methods at $\alpha = 0.05$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.933	0.949	0.876	0.934	1.336	0.664	0.152	0.000
	Split (B1/B2)	yes	0.933	0.949	0.876	0.936	1.335	0.665	0.151	0.000
	HCP (B3)	no	0.950	0.966	0.904	0.953	1.443	0.557	0.095	0.000
	HCP (B3)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.075	0.000
	Subsample once (B3)	no	0.950	0.965	0.902	0.952	1.441	0.559	0.098	0.000
	Subsample once (B3)	yes	0.963	0.967	0.951	0.950	1.515	0.485	0.076	0.000
	Repeated subsample (B3)	no	0.951	0.966	0.904	0.953	1.444	0.556	0.094	0.000
	Repeated subsample (B3)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.076	0.000
	GWCP (B4)	no	0.949	0.964	0.901	0.951	1.432	0.568	0.101	0.000
	GWCP (B4)	yes	0.961	0.966	0.950	0.948	1.507	0.493	0.080	0.000
	ProtCOVER (i)+(ii)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	ProtCOVER (i)+(ii)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	ProtCOVER (i)+(iii)	no	0.950	0.966	0.904	0.953	1.443	0.557	0.095	0.000
	ProtCOVER (i)+(iii)	yes	0.963	0.968	0.952	0.950	1.517	0.483	0.075	0.000
	ProtCOVER full	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	ProtCOVER full	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	Support-only abst. (B5)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	Support-only abst. (B5)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
	Random abst. (B5)	no	0.951	0.966	0.904	0.953	1.445	0.555	0.094	0.000
	Random abst. (B5)	yes	0.964	0.969	0.951	0.952	1.523	0.477	0.073	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.928	0.946	0.859	0.934	1.330	0.670	0.167	0.000
	Split (B1/B2)	yes	0.927	0.944	0.843	0.940	1.325	0.675	0.167	0.000
	HCP (B3)	no	0.946	0.962	0.889	0.950	1.431	0.569	0.108	0.000
	HCP (B3)	yes	0.960	0.967	0.936	0.952	1.516	0.484	0.080	0.000
	Subsample once (B3)	no	0.947	0.963	0.890	0.951	1.441	0.559	0.105	0.000
	Subsample once (B3)	yes	0.958	0.966	0.935	0.949	1.510	0.490	0.084	0.000
	Repeated subsample (B3)	no	0.946	0.962	0.889	0.950	1.432	0.568	0.109	0.000
	Repeated subsample (B3)	yes	0.960	0.967	0.935	0.952	1.515	0.485	0.080	0.000
	GWCP (B4)	no	0.947	0.964	0.892	0.951	1.447	0.553	0.104	0.000
	GWCP (B4)	yes	0.961	0.966	0.944	0.950	1.527	0.473	0.081	0.000
	ProtCOVER (i)+(ii)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	ProtCOVER (i)+(ii)	yes	0.965	0.969	0.951	0.955	1.553	0.447	0.069	0.000
	ProtCOVER (i)+(iii)	no	0.946	0.962	0.889	0.950	1.431	0.569	0.108	0.000
	ProtCOVER (i)+(iii)	yes	0.960	0.967	0.936	0.952	1.516	0.484	0.080	0.000
	ProtCOVER full	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	ProtCOVER full	yes	0.965	0.969	0.951	0.955	1.553	0.447	0.069	0.000
	Support-only abst. (B5)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	Support-only abst. (B5)	yes	0.965	0.969	0.951	0.955	1.554	0.446	0.069	0.000
	Random abst. (B5)	no	0.954	0.969	0.905	0.957	1.489	0.511	0.086	0.000
	Random abst. (B5)	yes	0.965	0.969	0.951	0.955	1.554	0.446	0.069	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.927	0.945	0.852	0.933	1.329	0.671	0.170	0.000
	Split (B1/B2)	yes	0.925	0.943	0.833	0.941	1.324	0.676	0.173	0.000
	HCP (B3)	no	0.944	0.961	0.884	0.949	1.427	0.573	0.114	0.000
	HCP (B3)	yes	0.957	0.966	0.929	0.950	1.512	0.488	0.087	0.000
	Subsample once (B3)	no	0.943	0.959	0.878	0.949	1.416	0.584	0.119	0.000
	Subsample once (B3)	yes	0.956	0.964	0.926	0.949	1.505	0.495	0.090	0.000
	Repeated subsample (B3)	no	0.945	0.961	0.883	0.949	1.427	0.573	0.115	0.000
	Repeated subsample (B3)	yes	0.957	0.966	0.929	0.949	1.511	0.489	0.087	0.000
	GWCP (B4)	no	0.946	0.963	0.888	0.950	1.456	0.544	0.109	0.000
	GWCP (B4)	yes	0.956	0.963	0.943	0.942	1.526	0.474	0.096	0.000
	ProtCOVER (i)+(ii)	no	0.963	0.975	0.926	0.964	1.580	0.420	0.068	0.000
	ProtCOVER (i)+(ii)	yes	0.969	0.974	0.962	0.958	1.616	0.384	0.064	0.000
	ProtCOVER (i)+(iii)	no	0.944	0.961	0.884	0.949	1.427	0.573	0.114	0.000
	ProtCOVER (i)+(iii)	yes	0.957	0.966	0.929	0.950	1.512	0.488	0.087	0.000
	ProtCOVER full	no	0.963	0.975	0.926	0.964	1.580	0.420	0.068	0.000
	ProtCOVER full	yes	0.969	0.974	0.962	0.959	1.617	0.383	0.063	0.014
	Support-only abst. (B5)	no	0.963	0.975	0.926	0.964	1.580	0.420	0.068	0.000
	Support-only abst. (B5)	yes	0.970	0.974	0.962	0.959	1.620	0.380	0.063	0.014
	Random abst. (B5)	no	0.963	0.975	0.926	0.964	1.580	0.420	0.068	0.000
	Random abst. (B5)	yes	0.970	0.974	0.962	0.959	1.620	0.380	0.063	0.014
S3 (orphan families)	Split (B1/B2)	no	0.929	0.944	0.870	0.931	1.330	0.670	0.159	0.000
	Split (B1/B2)	yes	0.930	0.944	0.865	0.936	1.330	0.670	0.156	0.000
	HCP (B3)	no	0.949	0.963	0.903	0.951	1.448	0.552	0.096	0.000
	HCP (B3)	yes	0.963	0.970	0.952	0.950	1.527	0.473	0.069	0.000
	Subsample once (B3)	no	0.946	0.961	0.899	0.948	1.440	0.560	0.105	0.000
	Subsample once (B3)	yes	0.962	0.970	0.952	0.948	1.525	0.475	0.070	0.000
	Repeated subsample (B3)	no	0.949	0.963	0.903	0.951	1.447	0.553	0.097	0.000
	Repeated subsample (B3)	yes	0.963	0.970	0.952	0.950	1.528	0.472	0.068	0.000
	GWCP (B4)	no	0.951	0.965	0.905	0.954	1.467	0.533	0.091	0.000
	GWCP (B4)	yes	0.964	0.969	0.951	0.953	1.527	0.473	0.069	0.000
	ProtCOVER (i)+(ii)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	ProtCOVER (i)+(ii)	yes	0.967	0.971	0.953	0.957	1.548	0.452	0.063	0.000
	ProtCOVER (i)+(iii)	no	0.949	0.963	0.903	0.951	1.448	0.552	0.096	0.000
	ProtCOVER (i)+(iii)	yes	0.963	0.970	0.952	0.950	1.527	0.473	0.069	0.000
	ProtCOVER full	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	ProtCOVER full	yes	0.967	0.971	0.953	0.958	1.548	0.452	0.063	0.000
	Support-only abst. (B5)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	Support-only abst. (B5)	yes	0.967	0.971	0.953	0.957	1.548	0.452	0.063	0.000
	Random abst. (B5)	no	0.956	0.969	0.912	0.959	1.499	0.501	0.078	0.000
	Random abst. (B5)	yes	0.967	0.971	0.953	0.957	1.548	0.452	0.063	0.000

Table 6. All methods at $\alpha = 0.2$, mean over predictors and seeds.

Setting	Method	cc	Cov	var.-w.	Cov_1	Cov_0	size	single.	under-cov.	abst.
S1 (exchangeable)	Split (B1/B2)	no	0.786	0.797	0.675	0.787	0.927	0.919	0.282	0.000
	Split (B1/B2)	yes	0.782	0.796	0.680	0.779	0.923	0.921	0.287	0.000
	HCP (B3)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	HCP (B3)	yes	0.840	0.846	0.804	0.803	1.045	0.952	0.164	0.000
	Subsample once (B3)	no	0.800	0.812	0.692	0.803	0.954	0.947	0.243	0.000
	Subsample once (B3)	yes	0.840	0.846	0.802	0.804	1.046	0.951	0.165	0.000
	Repeated subsample (B3)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	Repeated subsample (B3)	yes	0.840	0.847	0.804	0.803	1.045	0.952	0.164	0.000
	GWCP (B4)	no	0.799	0.810	0.692	0.800	0.949	0.947	0.246	0.000
	GWCP (B4)	yes	0.839	0.845	0.803	0.801	1.043	0.953	0.166	0.000
	ProtCOVER (i)+(ii)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.000
	ProtCOVER (i)+(ii)	yes	0.840	0.847	0.799	0.805	1.044	0.953	0.163	0.000
	ProtCOVER (i)+(iii)	no	0.800	0.812	0.694	0.802	0.953	0.949	0.242	0.000
	ProtCOVER (i)+(iii)	yes	0.840	0.847	0.804	0.803	1.046	0.951	0.164	0.001
	ProtCOVER full	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	ProtCOVER full	yes	0.840	0.847	0.800	0.806	1.044	0.953	0.163	0.000
	Support-only abst. (B5)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	Support-only abst. (B5)	yes	0.840	0.847	0.799	0.806	1.044	0.953	0.163	0.000
	Random abst. (B5)	no	0.800	0.812	0.693	0.802	0.953	0.949	0.242	0.001
	Random abst. (B5)	yes	0.840	0.847	0.799	0.806	1.045	0.953	0.163	0.000
S2, $\beta = 1$	Split (B1/B2)	no	0.777	0.789	0.641	0.788	0.920	0.909	0.305	0.000
	Split (B1/B2)	yes	0.774	0.787	0.630	0.792	0.917	0.914	0.308	0.000
	HCP (B3)	no	0.790	0.803	0.658	0.801	0.942	0.941	0.266	0.000
	HCP (B3)	yes	0.829	0.836	0.754	0.811	1.024	0.967	0.187	0.000
	Subsample once (B3)	no	0.786	0.798	0.653	0.798	0.935	0.933	0.280	0.000
	Subsample once (B3)	yes	0.828	0.836	0.752	0.810	1.023	0.965	0.189	0.000
	Repeated subsample (B3)	no	0.790	0.803	0.659	0.801	0.943	0.941	0.266	0.000
	Repeated subsample (B3)	yes	0.828	0.836	0.754	0.810	1.023	0.967	0.189	0.000
	GWCP (B4)	no	0.794	0.806	0.663	0.804	0.948	0.943	0.259	0.000
	GWCP (B4)	yes	0.835	0.842	0.785	0.804	1.050	0.944	0.179	0.000
	ProtCOVER (i)+(ii)	no	0.800	0.813	0.672	0.810	0.960	0.950	0.246	0.000
	ProtCOVER (i)+(ii)	yes	0.844	0.850	0.800	0.810	1.075	0.921	0.162	0.000
	ProtCOVER (i)+(iii)	no	0.790	0.803	0.658	0.801	0.942	0.941	0.266	0.000
	ProtCOVER (i)+(iii)	yes	0.829	0.836	0.754	0.811	1.025	0.967	0.187	0.001
	ProtCOVER full	no	0.800	0.813	0.673	0.811	0.962	0.949	0.246	0.001
	ProtCOVER full	yes	0.844	0.850	0.801	0.811	1.075	0.921	0.162	0.000
	Support-only abst. (B5)	no	0.800	0.813	0.672	0.811	0.962	0.949	0.246	0.001
	Support-only abst. (B5)	yes	0.844	0.850	0.801	0.810	1.075	0.921	0.162	0.000
	Random abst. (B5)	no	0.800	0.813	0.673	0.811	0.962	0.949	0.246	0.001
	Random abst. (B5)	yes	0.844	0.850	0.801	0.810	1.075	0.921	0.162	0.000
S2, $\beta = 2$	Split (B1/B2)	no	0.776	0.788	0.633	0.789	0.920	0.907	0.306	0.000
	Split (B1/B2)	yes	0.773	0.786	0.618	0.794	0.917	0.913	0.312	0.000
	HCP (B3)	no	0.790	0.803	0.650	0.803	0.942	0.941	0.268	0.000
	HCP (B3)	yes	0.824	0.833	0.736	0.812	1.018	0.971	0.197	0.000
	Subsample once (B3)	no	0.788	0.801	0.648	0.801	0.939	0.937	0.274	0.000
	Subsample once (B3)	yes	0.825	0.834	0.736	0.813	1.020	0.967	0.196	0.000
	Repeated subsample (B3)	no	0.791	0.803	0.650	0.803	0.942	0.941	0.266	0.000
	Repeated subsample (B3)	yes	0.824	0.833	0.736	0.811	1.018	0.970	0.197	0.000
	GWCP (B4)	no	0.798	0.810	0.661	0.810	0.957	0.941	0.254	0.000
	GWCP (B4)	yes	0.830	0.837	0.778	0.798	1.054	0.931	0.192	0.000
	ProtCOVER (i)+(ii)	no	0.808	0.822	0.676	0.819	0.980	0.945	0.228	0.000
	ProtCOVER (i)+(ii)	yes	0.843	0.850	0.796	0.811	1.081	0.912	0.165	0.000
	ProtCOVER (i)+(iii)	no	0.790	0.803	0.650	0.803	0.942	0.941	0.268	0.000
	ProtCOVER (i)+(iii)	yes	0.825	0.833	0.736	0.812	1.019	0.970	0.196	0.001
	ProtCOVER full	no	0.812	0.825	0.685	0.822	0.999	0.927	0.224	0.018
	ProtCOVER full	yes	0.846	0.854	0.801	0.814	1.101	0.892	0.160	0.023
	Support-only abst. (B5)	no	0.812	0.825	0.683	0.822	0.999	0.928	0.224	0.018
	Support-only abst. (B5)	yes	0.846	0.853	0.800	0.814	1.098	0.895	0.161	0.023
	Random abst. (B5)	no	0.812	0.825	0.683	0.822	1.000	0.927	0.224	0.018
	Random abst. (B5)	yes	0.846	0.853	0.800	0.814	1.100	0.893	0.161	0.023
S3 (orphan families)	Split (B1/B2)	no	0.778	0.789	0.664	0.781	0.924	0.912	0.300	0.000
	Split (B1/B2)	yes	0.774	0.788	0.664	0.775	0.919	0.916	0.307	0.000
	HCP (B3)	no	0.794	0.805	0.685	0.797	0.951	0.946	0.257	0.000
	HCP (B3)	yes	0.836	0.846	0.799	0.796	1.040	0.955	0.174	0.000
	Subsample once (B3)	no	0.795	0.806	0.682	0.801	0.954	0.943	0.255	0.000
	Subsample once (B3)	yes	0.835	0.846	0.799	0.794	1.039	0.958	0.176	0.000
	Repeated subsample (B3)	no	0.794	0.806	0.686	0.797	0.951	0.946	0.257	0.000
	Repeated subsample (B3)	yes	0.836	0.846	0.798	0.796	1.040	0.956	0.175	0.000
	GWCP (B4)	no	0.802	0.814	0.695	0.806	0.967	0.957	0.236	0.000
	GWCP (B4)	yes	0.839	0.847	0.792	0.805	1.046	0.948	0.170	0.000
	ProtCOVER (i)+(ii)	no	0.807	0.819	0.700	0.810	0.976	0.956	0.227	0.000
	ProtCOVER (i)+(ii)	yes	0.843	0.850	0.793	0.812	1.056	0.939	0.162	0.000
	ProtCOVER (i)+(iii)	no	0.794	0.805	0.685	0.797	0.951	0.946	0.257	0.000
	ProtCOVER (i)+(iii)	yes	0.837	0.847	0.800	0.797	1.044	0.951	0.173	0.005
	ProtCOVER full	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	ProtCOVER full	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003
	Support-only abst. (B5)	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	Support-only abst. (B5)	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003
	Random abst. (B5)	no	0.807	0.819	0.701	0.810	0.977	0.955	0.227	0.001
	Random abst. (B5)	yes	0.843	0.850	0.793	0.812	1.058	0.937	0.162	0.003